

CIRCUMFERENCE

Lesson 1: Around Circumference

Australian Curriculum: Mathematics (Year 8)

ACMMG197: Investigate the relationship between features of circles such as circumference, area, radius and diameter. Use formulas to solve problems involving circumference and area.

Lesson abstract

Students investigate whether the height of a cylinder containing three tennis balls is greater than its circumference. They ask what would happen if the balls were smaller or larger, and conclude that it is always the case that the circumference is a little more than three times the diameter or six times the radius. They apply this to estimations relating to real world circular objects.

Mathematical purpose (for students)

The ratio of circumference to diameter of any circle is “a bit more than three”.

Mathematical purpose (for teachers)

The lesson examines of the relationship between the circumference and diameter of a circle. The ratio of circumference to diameter is identified as a little more than 3. The number π is used, with the promise that a more accurate value will be found in subsequent lessons. It is intended that by using three-dimensional objects, students will be able to better relate concepts usually presented only in two-dimensions to real world objects.

Circumference and diameter lengths are directly compared, rather than comparing measurements only, to focus on the relationship rather than the numbers. The lesson involves students by asking them to publically commit to opinions before data is collected, by manipulating physical objects and by promoting discussion using group work.

At the end of this lesson, students will be able to:

- Approximate the *circumference* given *radius* or *diameter*.
- Approximate the *diameter* or *radius* given the *circumference*.

Lesson Length 60 minutes approximately

Vocabulary Encountered Lesson Materials

- | | |
|-----------------|---|
| • centre | • 3 x tennis balls per group |
| • circle | • 3 x A4 and 1 x A3 sheets of paper per group |
| • circumference | • 3-4 different spheres/balls larger than a tennis ball (e.g. netball) |
| • diameter | • 3-4 different spheres/balls smaller than a tennis ball (e.g. squash ball) |
| • great circle | • images for printing or screening (1a Circumference Images PowerPoint) |
| • perimeter | • 1b Circumference Consolidation PowerPoint |
| • radius | |
| • sphere | |

We value your feedback after this lesson via <http://tiny.cc/lesson-feedback>



Getting Started

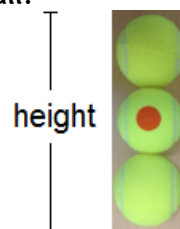
Show students the image of a stack of 3 tennis balls (1a Circumference Images PowerPoint) and ask:

“Is the height of this stack more or less than the circumference of the single ball?”



circumference

Students show whether they think that the height of the stack is more or less than the circumference of a single ball by moving to either side of the classroom.



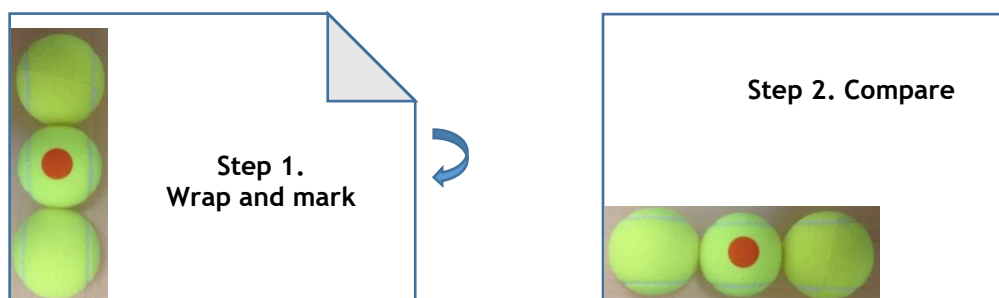
Ask students to think of a way to check the result without using a ruler or tape measure. Encourage students to think about **comparing**, **without actually measuring**, the two dimensions. This puts the emphasis on the ratio, rather than calculations.

Initial Task - Tennis Balls

Students work in groups of 3-4 to answer the question above.

Suggested comparison strategy

1. **Wrap** the A4 paper around the set of tennis balls and mark the circumference;
2. **Unwrap** and **compare** the marked circumference with the overall height of the three tennis balls



Expected Student Response

Students doing the comparison correctly will find that the circumference is just a little longer than the height of the stack of three balls. They may also suggest that the circumference would be expected to be considerably shorter than the height of the stack of four balls.

Extending prompt

Wilson, the makers of the official tennis ball used in the Australian Open, sell packs of four tennis balls in a cylinder.

- Will the height of Wilson’s cylinders be more or less than the circumference of one tennis ball?

Class discussion

Which was larger: the circumference of one tennis ball or the overall height of the stack of three tennis balls?

During this discussion, identify clearly what lengths have been compared, and what is meant by the *circumference*, *radius* and *diameter* of a sphere, such as a tennis ball.

- Identify the largest circle on the surface of a tennis ball (a *great circle*).
- Discuss also how an infinite number of these exist on the surface of a single tennis ball and where their centres are. It could be helpful to use the analogy of putting a belt around the equator, and taking a journey to the centre of the earth.

Establish the link between the circumference of a sphere and the circumference of a circle. Show the image of half of a tennis ball to the class (1a Circumference Images PowerPoint).

- The *perimeter* of a 2D circle is called its *circumference* and all the points on the circumference of a circle are the same distance from the centre.
- Terms *radius* and *diameter* should be defined.
- A labelled diagram should be recorded in each student's notebook.

The Task

Students test whether their conclusions hold if a ball larger or smaller than a tennis ball is used. Clarifying what may happen if the object is bigger or smaller than a tennis ball encourages deeper consideration of spherical and circular properties.

Students work in groups of 3-4 to determine their solution to:

“Is the height of ALL 3 ball stacks more or less than the circumference of a single ball?”

Suggested comparison strategy

1. **Wrap** the paper around a different sized ball to get a length on the paper equal to the circumference; use the same approach as used for the tennis ball.
2. **Unwrap and compare** the marked length with the overall height of three of the new balls. If only one ball is available, measure the diameter using a wall and book:
 - I. Mark the distance.
 - II. Move the paper towards the wall and reset from the recently marked distance.
 - III. Repeat steps 1 and 2 for a total of three diameters.



Expected Student Response

Most students will find that the circumference is again just a little larger than the height of a stack of three of this different sized ball. They are likely to suggest that the size of the spheres doesn't matter.

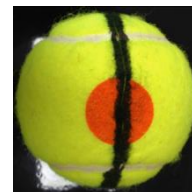
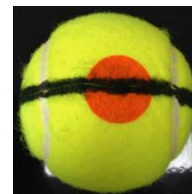
The key idea here is that all circles are similar, so the ratio of circumference to diameter is independent of size.

Reflection

What did you notice about the relationship between the circumference and the diameter of the balls? What impact did the size of the ball have?

Show students the images of oblate solids rather than spheres (1a Circumference Images PowerPoint) and ask:

- What if we used eggs, AFL or rugby balls? Would the height of a stack of three of these be more or less than the distance around the middle of one?
- What is special about a sphere?



Expected Student Response

- It depends on which way around these different objects are - up tall or down flat.
- Spheres and circles are special because the circumference and diameter are the same no matter where you measure.
- The circumference of a sphere or circle is ALWAYS a little more than three times the diameter.
- For all spheres or circles, circumference = “3 and a bit” times diameter.

Introduce π

We can also express this relationship by saying that the ratio of circumference to diameter is a constant number regardless of the size of the sphere or circle. We write this number as π , which is said ‘pi’. Hence we can write $C = \pi d$. In the next lesson we will find a value for π that is better than “3 and a bit”.

The final two dot points above should be written in student workbooks as well as the symbol π and its meaning, and the formula $C = \pi d$.

Applications and Consolidation

In this final stage, students view a series of photos of real world circular objects (1b Circumference Consolidation PowerPoint) and are asked a question related to the circumference. Each object has 3 slides with progressively more information on each slide. Students are given the opportunity to adjust their answer after each slide.

Move fairly quickly through the three slides:

1. Slide 1 describes a scenario
2. Slide 2 provides limited visual information
3. Slide 3 provides a frame of reference

Teacher Notes

- Students should prepare a sheet of paper (from notebook or folder) with “greater than” and $>$ symbol on one side and “less than” and $<$ on the other.
- They respond to the first slide of each set by holding up the symbol that reflects their estimation.
- When new information becomes available on subsequent slides they can make a better estimation and if desired can change their symbol.
- Alternatively, students can write the $>$ sign on one page of their workbooks and just hold the workbook upside down for $<$.